An Internship Report

Of

Python Full Stack Development with Project

Submitted

To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech (Computer Science and Engineering)

By

Roll Number of Student: 2410031253

Name of Student: ANSHU RANA

Batch: 2024-2028

2026-27

CANDIDATE'S DECLARATION

I, ANSHU RANA, do hereby declare that the internship report titled “Python Full Stack Development with Project” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:-
Student Name: ANSHU RANA
Roll No.: 2410031253

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluably constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

Signature:-

Student Name: ANSHU RANA

Roll No.: 2410031253

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3. INTERNSHIP COMPLETION CERTIFICATE

शिक्षा मंत्रालय
MINISTRY OF EDUCATION

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All India Council for Technical Education

NATIONAL INTERNSHIP PORTAL

EduSkills®
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AICTE - EduSkills 2026 VIRTUAL INTERNSHIP

Certificate of Virtual Internship

This is to certify that

Anshu rana

IILM University, Greater Noida

has successfully completed the 8-weeks

Python Full Stack Development With Project Virtual Internship

During June - August 2026

Supported by EduSkills ACADEMY

Dr. Buddha Chandrasekhar
Chief Coordinating Officer (CCO)
AICTE, Ministry of Education

Shubhajit Jagadev
Chief Executive Officer (CEO)
EduSkills

Certificate ID : 4ff6a35af86f89354446

Student ID: STU69ecf30a46a981777136394

INTERNSHIP GRADE E

GRADE- O (Outstanding):90-100 | E (Excellent):80-89 | A (Very Good):70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

Internship Offer Letter

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Ref No: C17Y2026M091494328

Internship Offer Letter

Student Name	: Anshu rana
Stream	: B.Tech
Branch	: B.Tech(Computer Science and Engineering)
Institute Name	: IILM University, Greater Noida
Internship Domain	: Python Full Stack Development + Project
Internship Duration	: June 2026 - August 2026
AICTE Student ID	: STU69ecf30a46a981777136394

We are pleased to inform you that you have been selected for the 8-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the 8-week internship in Python Full Stack Development with Project. The internship provided practical exposure to the development of full-stack web applications, covering both frontend and backend development, database integration, API development, testing, and deployment.

The internship focused on Python for backend development, FastAPI for APIs, Pydantic for data validation, React for frontend development, and SQLAlchemy for database integration. It also covered React Hooks, state management, routing, forms, API design, and performance optimization.

As part of the internship, practical project work included an E-commerce Platform with User Authentication and Product Catalog and a Basic Authentication Simulation.

4.2 Organization Profile

The internship program provided structured training in Python Full Stack Development, combining theoretical learning with practical project work.

The program focused on modern web development technologies including Python, FastAPI, React, Pydantic, and SQLAlchemy. It provided exposure to the complete process of developing a web application, from project setup and backend development to frontend integration, testing, and deployment.

4.3 Problem Statement

Developing modern web applications requires knowledge of multiple technologies and their integration. Learning frontend, backend, databases, and APIs separately may not provide sufficient understanding of how a complete application works.

The internship addressed this by providing practical training in Python backend development, FastAPI API creation, React frontend development, database integration, and frontend-backend communication. It also introduced testing, deployment, API design, and performance optimization.

4.4 Project Objectives

The main objectives of the internship were:

- To understand the fundamentals of full-stack development.
- To strengthen Python programming skills for backend development.
- To develop APIs using FastAPI.
- To understand data validation using Pydantic.
- To learn React and frontend development.
- To understand React Hooks and state management.
- To create reusable React components.
- To integrate React with FastAPI.
- To understand database integration using SQLAlchemy.
- To learn routing, forms, and user-input handling.
- To understand API design, testing, and deployment.
- To apply the learned concepts through practical projects.

4.5 Scope of the Project

The scope of the internship covered Python-based backend development and React-based frontend development. Backend topics included Python, FastAPI, Pydantic, API development, and SQLAlchemy database integration.

Frontend topics included React, Hooks, reusable components, state management, routing, navigation, forms, and styling. The internship also covered frontend-backend integration, testing, deployment, API design, and performance optimization.

The practical scope included an E-commerce Platform with User Authentication and Product Catalog and a Basic Authentication Simulation

4.6 Technologies and Tools Used

- Programming Language: Python
- Backend Development: FastAPI, Pydantic
- Frontend Development: React, React Hooks, Reusable React Components
- State Management: Redux, Zustand
- Database Integration: SQLAlchemy
- API Development: FastAPI APIs, API Design Principles and Best Practices
- Frontend Development Tools: React Routing, Navigation, Forms, Styling
- Testing & Deployment: Full-Stack Application Testing, Deployment Strategies
- Projects: E-commerce Platform with User Authentication and Product Catalog, Basic Authentication Simulation

4.7 System Architecture

The general architecture followed during the internship for the Python Full Stack learning track can be summarized in the following layers:

- Frontend Layer: React provides the user interface with reusable components, Hooks, routing, forms, and state management.
- API Layer: FastAPI handles backend APIs and communication between the frontend and backend.
- Validation Layer: Pydantic validates and models the data exchanged through APIs.
- Database Layer: SQLAlchemy provides database integration for storing and managing application data.
- Integration Layer: React and FastAPI are integrated to enable smooth frontend-backend communication.
- Testing & Deployment Layer: The application is tested and prepared for deployment using suitable deployment strategies.
- Application Layer: These technologies were applied in projects including an E-commerce Platform and Basic Authentication Simulation.
- Assessment Layer: Weekly quizzes, learning-pathway badges, and a Final Credential Validation / Internship Grade Point Assessment track progress and final grading.

4.8 Methodology

The internship followed a structured, week-wise methodology combining conceptual learning, hands-on labs, and quiz-based assessments, culminating in a Final Credential Validation, as summarized in the table below.

Week	Module	Description
Week 1	Introduction to Full Stack & Python Fundamentals	Introduction to full-stack development concepts and environment setup. Learn
Week 2	FastAPI & Data Validation (Pydantic)	Build backend APIs using FastAPI framework. Learn request handling, routing,
Week 3	React Fundamentals & Hooks	Learn React basics including components, JSX, props, state management,
Week 4	Reusable Components & Frontend-Backend Integration	Create reusable React components and integrate frontend with FastAPI
Week 5	Database Integration with SQLAlchemy	Integrate databases using SQLAlchemy ORM, perform CRUD operations, manage
Week 6	Advanced React & State Management	Learn advanced React patterns, routing with React Router, and state
Week 7	Testing & Deployment	Implement backend and frontend testing strategies. Learn deployment
Week 8	API design & Front end optimisation Capstone Project – Full Stack	Begin building a complete full-stack application (E-commerce platform)

Note: There was no stipend associated with this internship.

Each week concluded with quiz-based learning-pathway assessments and skill badges to evaluate conceptual understanding. In the final phase, a Final Credential Validation / Internship Grade Point Assessment was conducted, and the overall grade was based on performance across the weekly modules and this final assessment.

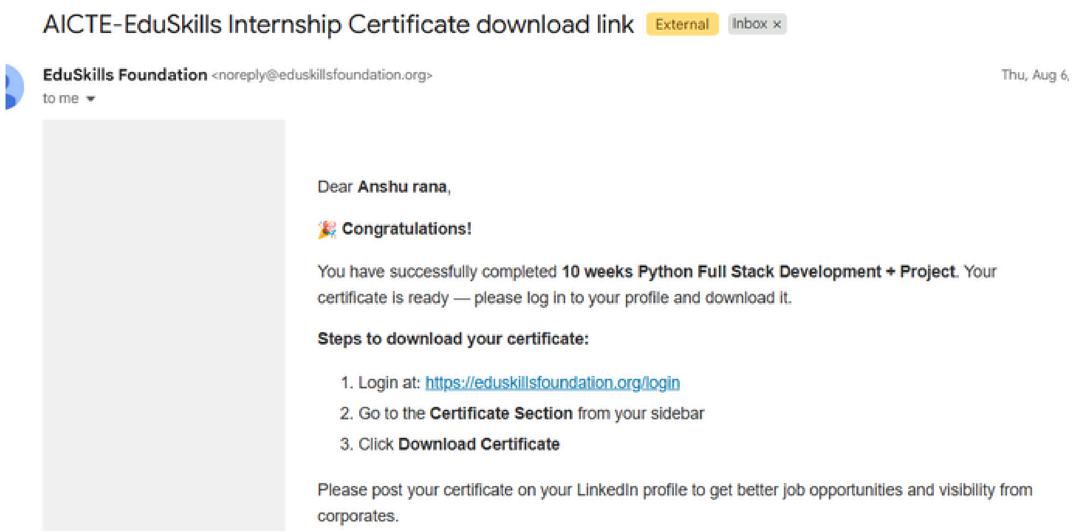
4.9 Expected Outcomes

The internship provided practical knowledge of:

- Python backend development
- FastAPI API development
- Pydantic data validation
- React frontend development
- React Hooks and state management
- Database integration using SQLAlchemy
- Frontend-backend integration
- Routing and form handling
- API design and testing
- Deployment concepts
- Full-stack project development

4.10 Certificates of Completion and Communication Proof

The internship was successfully completed as an 8-week program in Python Full Stack Development with Project. The certificate confirms successful completion of the internship requirements and project work are attached in Chapter 3 of this report. The official completion e-mail from EduSkills Foundation and the corresponding skill badges earned on the Python Full Stack Development Program profile are attached below as further communication proof.



AICTE-EduSkills Internship Certificate download link External Inbox ×

EduSkills Foundation <noreply@eduskillsfoundation.org> Thu, Aug 6,

to me

Dear **Anshu rana**,

🎉 **Congratulations!**

You have successfully completed **10 weeks Python Full Stack Development + Project**. Your certificate is ready — please log in to your profile and download it.

Steps to download your certificate:

1. Login at: https://eduskillsfoundation.org/login
2. Go to the **Certificate Section** from your sidebar
3. Click **Download Certificate**

Please post your certificate on your LinkedIn profile to get better job opportunities and visibility from corporates.

5. BIBLIOGRAPHY/REFERENCES

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6. SQLAlchemy Documentation, https://www.sqlalchemy.org/
7. Redux Documentation, https://redux.js.org/
8. Zustand Documentation, https://zustand.docs.pmnd.rs/